Practice Set 6 Paper 4 – Recent Advances

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Practice paper compiled from the question bank.

Q1. Present concept in management of hypospadias

Type: LAQ (25 marks) (embryology and proximal repair also in 2017-Winter Paper 3, Q1; Snodgrass repair in 2023-Winter Paper 3, Q3c)

Read as "present (current) concepts in the management of hypospadias".

Introduction

- Hypospadias is an arrest of the androgen-dependent "closing zipper" of urethral fold fusion (weeks 8–16). Triad: ventral ectopic meatus, ventral curvature, dorsal hooded prepuce.
- 1 in 150–300 male births; distal 70–85%, proximal 10–25%.
- **Goals:** a straight erection, a urethra to the tip (standing voiding, semen delivery) and acceptable appearance, with the fewest operations and a result lasting into adulthood.

Concept 1: severity is judged in theatre, not by the meatus

- A "distal" meatus may hide absent spongiosum and severe curvature, so classify **after degloving**.

GMS (glans–meatus–shaft) score	Glans	Meatus	Shaft curvature
1	Good size, deep groove	Glanular	None
2	Adequate, grooved	Coronal	<30°
3	Small; narrow or flat plate	Mid or distal shaft	30–60°
4	Very small; flat plate	Proximal shaft, penoscrotal	>60°

- Total 3–12; a higher score predicts more complications.
- **POST (plate objective scoring tool):** a low ratio of neomeatus length to glans fusion length means a poor plate, which favours an inlay graft over plain TIP (tubularised incised plate).

Concept 2: evaluation

- **DSD (disorder of sex development) work-up** if hypospadias plus undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone).
- No routine VCUG (voiding cystourethrogram) or renal ultrasound. Associations: hernia or hydrocele 9–16%, cryptorchidism about 7% (10% in proximal), enlarged utricle.
- **Aetiology:** causes converge on deficient androgen action (placental insufficiency, assisted reproduction, genes, endocrine disruptors).

Concept 3: timing and preparation

- Repair at **6–18 months** (AAP (American Academy of Pediatrics): 6–12 months), before genital awareness at about 18 months. One short anaesthetic appears low-risk.
- A straight penis that voids standing may gain little from surgery: counsel honestly.
- **Preoperative androgen** if glans width <14–15 mm: testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Evidence mixed; two RCTs (randomised controlled trials) showed fewer complications.

Concept 4: curvature decides the operation

```
Deglove + excise dysplastic dartos + artificial erection
  ├─ straight       → urethroplasty
  ├─ residual <30°  → dorsal midline plication (12 o'clock nerve-free zone)
  └─ >30°           → divide plate → still >30° → ventral corporotomy ± graft → staged repair
```

- Recurrence 28% after plication versus 9% after grafting. Under-correction causes repeated surgery, so correct fully first time.

Concept 5: choice of technique

Situation	Current choice
Glanular meatus, no curvature	Advancement: MAGPI (meatal advancement and glanuloplasty), MIV (M inverted V) glansplasty; selectively
Distal or midshaft, good plate	**TIP (Snodgrass)**: >90% of distal, about 80% of midshaft repairs; Thiersch–Duplay; GAP (glans approximation procedure) for a deep groove
Narrow or shallow plate	TIP with **dorsal inlay graft**; onlay island flap; Mathieu (SLAM modification)
Proximal, curvature >30°	**Two-stage**: STAG (staged tubularised autograft) or Bracka graft; Byars flaps; distal-first (Ulaanbaatar)
Proximal, curvature corrected, good tissue	One-stage onlay or TPIF (transverse preputial island flap); proximal TIP abandoned

- **Why stage:** a urethroplasty laid over a fresh corporal graft fails more often; staging gives a supple, vascular bed. Stage 2 after ≥6 months.

Concept 6: protect blood supply

- Magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), rapidly absorbed fine sutures, no hair-bearing skin, haemostasis.
- Postoperative: stent 3–21 days by complexity, compressive dressing, oxybutynin; antibiotic prophylaxis debated.

Concept 7: standardised complications and outcomes

- Definitions: glans dehiscence (fusion <2 mm); meatal stenosis (symptoms plus <8 Fr before puberty, <12 Fr after); recurrent curvature >30°.
- TIP meta-analysis (Pfistermuller 2015): complications 10.6%, fistula 5.7%. Proximal complications 23–68%; STAG about 23% (lowest); Byars 43–81%.
- A long, rigid, spongiosum-free neourethra raises resistance (Poiseuille's law), causing fistula and diverticulum. Fistula: see 2015-Summer Paper 4, Q3a.
- **Also know:** BXO (balanitis xerotica obliterans), excise and stage with buccal graft; redo repairs use buccal (lip inside glans) mucosa, Cecil scrotal parasitisation, tissue expansion.

Concept 8: follow-up and patient-reported outcomes

- About three-quarters of complications appear by 3 years; stricture and curvature appear late. Follow proximal cases **through puberty**.
- Tools: HOSE (Hypospadias Objective Scoring Evaluation, surgeon), PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation, photo panels). Decision aids reduce parental regret.

Recent advances

- Registries, AI (artificial intelligence) video review, genomic risk profiling, a quality-of-life measure, engineered tissue.

Diagram to draw: curvature algorithm and two-stage graft repair.

Key points to remember:

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.

Q2. Discuss recent advances in tissue engineering with respect to urology

Type: LAQ (25 marks) (also answered in 2023-Winter Paper 1, Q1 and 2022-Summer Paper 4, Q2; this answer stresses recent advances)

Definition and rationale

- **Tissue engineering** applies engineering and life sciences to build biological substitutes that restore, maintain or improve tissue function (Langer and Vacanti, 1993).
- **Why in urology:** bowel in the urinary tract causes mucus, stones, metabolic acidosis, infection and cancer; grafts for long urethral defects are scarce. Aim: replace "wrong tissue" with "right tissue".

Basic principle (the triad)

```
Biopsy → cells expanded in culture → seeded on scaffold (+ signals)
       → conditioned in bioreactor → implanted → host vascularises and remodels
```

- **Unseeded** matrices depend on host cells migrating in, so they suit only short onlay defects (about 0.5 cm for tubes in animal work). Longer or tubular gaps need **seeded** scaffolds, or they fibrose and contract.

Recent advances

1. Cell sources

Source	Advance and limitation
Autologous urothelial and smooth muscle cells	Standard; unusable if the bladder is malignant
Adult stem cells: bone marrow or adipose mesenchymal, muscle-derived, **urine-derived**	Easy, non-invasive harvest; act mainly by paracrine growth factors
iPSCs (induced pluripotent stem cells)	Adult cells reprogrammed; patient-specific, no ethical issue; tumour risk
Embryonic stem cells	Pluripotent; ethics, teratoma risk

- Gene-edited and gene-modified cells are being explored. Cell sheet engineering builds layers without a scaffold.

2. Scaffolds

- Requirements: biocompatible, degrade as tissue forms, porous for cells and vessels, mechanically suited (compliant bladder, non-collapsing tube).

Type	Examples
Acellular matrices	Bladder acellular matrix, SIS (small intestinal submucosa), decellularised dermis: keep native architecture and growth factors
Synthetic polymers	PGA (polyglycolic acid), PLA (polylactic acid), PLGA (copolymer): controlled shape and degradation
Newer	Electrospun nanofibres and hydrogels mimicking extracellular matrix; **"smart" scaffolds** releasing growth factors such as VEGF (vascular endothelial growth factor); silk fibroin

3. Bioreactors

- Perfuse nutrients through thick constructs and apply **cyclic stretch and filling**, so engineered bladder muscle is conditioned before implantation.

4. Solving vascularity (the main barrier)

- Oxygen diffuses only about 100–200 µm, so thick constructs die centrally.
- Strategies: omental wrap (Atala), angiogenic factors, co-seeding endothelial cells, printed vascular channels.

5. 3D bioprinting

- Cells in hydrogel **"bio-ink"** laid layer by layer from CT (computed tomography) or MRI (magnetic resonance imaging) data, placing several cell types precisely with built-in channels.
- Bioprinted urethra and bladder constructs are at the animal stage. Printed non-living kidney and prostate models already aid planning and teaching.

6. Organoids

- Self-organising 3D mini-organs: **kidney organoids** from iPSCs model disease and drug toxicity; **bladder and prostate tumour organoids** allow personalised drug testing.

7. Whole-organ decellularisation

- Kidneys are stripped of cells by detergent perfusion, leaving the matrix and vascular tree, then recellularised with the patient's cells; still at the animal stage. Renal assist devices are a parallel route.

Clinical applications: where we stand

Organ	Status
Bladder	**Atala (2006):** seeded collagen–PGA construct with omental wrap in **7 myelomeningocele patients**, compliance improved. Industry phase II trial: **no significant gain in compliance or capacity**. Still experimental (Chapter 48)
Conduit	Neo-Urinary Conduit trial (NCT01087687): survived about 250 days on average; **half developed stenosis or stricture**; stiff material, no blood supply (Chapter 135)
Urethra	Acellular matrix onlay for short strictures; **Raya-Rivera (2011):** seeded tubular PGA–PLGA urethras in **5 boys**, functional up to 6 years; tissue-engineered oral mucosa avoids donor-site morbidity
Sphincter, detrusor	Autologous muscle-derived cells for SUI (stress urinary incontinence); for underactive detrusor, open-label studies improved voiding efficiency (Chapter 111)
Vagina	Engineered vaginas in MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome (Raya-Rivera, 2014); off-the-shelf SIS grafts (Chapter 59)
Penis, testis	Corporal tissue (animal); stem cells for ED (erectile dysfunction) investigational; spermatogonial stem cell banking

- An engineered bladder gives **storage only**: innervation for voluntary emptying cannot yet be recreated (Chapter 111).
- **Also know:** robotic retrieval of a **vascularised composite bladder allograft** (Chapter 135) is a transplant, not engineered, route to "right tissue", needing immunosuppression.

Challenges

- Vascularity, innervation, contractility; fibrosis and contracture.
- Scale-up, cost, culture time, regulation; stem cell tumour risk.
- Promising pilots followed by disappointing controlled trials.

Future

- Vascularised bioprinted tissue, off-the-shelf cell-free smart scaffolds, combined gene and cell therapy.

Diagram to draw: biopsy → culture → seeding on scaffold → bioreactor → implant with omental wrap.

> **Key points to remember:**
> - Triad: cells, scaffold, signals; vascularity (100–200 µm diffusion) is the main barrier.
> - Atala 2006 promising; phase II trial no gain; conduit trial half stenosed.
> - Raya-Rivera 2011: seeded urethras in 5 boys.
> - New: iPSCs, urine-derived cells, bioprinting, organoids.

Q3a. 3D printing in urology

Type: SAQ (10 marks)

Definition

- **3D (three-dimensional) printing** is additive manufacturing: a solid object is built layer by layer from a digital model.
- In urology it turns a patient's CT (computed tomography) or MRI (magnetic resonance imaging) into a physical replica of kidney, tumour, vessels or prostate.

Workflow

```
Thin-slice CT/MRI (DICOM files) → segmentation of kidney, tumour, vessels, calyces
   → STL (standard tessellation language) mesh → printer → post-processing
```

- Takes hours to days; segmentation needs expertise.

Printing technologies

Method	Principle	Use
FDM (fused deposition modelling)	Melted plastic filament extruded	Cheap, rigid models
SLA (stereolithography)	Laser cures liquid resin	Fine detail, transparent models
Material jetting (PolyJet)	Jetted photopolymers, multi-colour	Tumour, vessels and calyces in different colours
SLS (selective laser sintering)	Laser fuses powder	Durable parts, devices
Bioprinting	Living cells in hydrogel "bio-ink"	Tissue engineering (see Practice Set 6 Paper 4, Q2)

Applications

1. **Partial nephrectomy planning:** complex hilar or endophytic tumours. The model shows the tumour against vessels and collecting system, so the surgeon plans the approach and rehearses steps. Promising.
2. **PCNL (percutaneous nephrolithotomy) planning:** staghorn stones, anomalous kidneys; choice of calyx. Early promise; 3D reformatting did not predict stone-free rate better.
3. **Prostate:** models with MRI lesions for margin and nerve-sparing planning and focal therapy.
4. **Patient education and consent:** patients understand their anatomy and the operation better.
5. **Training and simulation:** silicone or hydrogel phantoms for PCNL puncture, partial nephrectomy, pyeloplasty, anastomosis.
6. **Patient-specific devices:** prototype stents, catheters and surgical guides.

Limitations

- Cost, printing time and segmentation errors.
- Materials do not reproduce tissue feel or bleeding.
- Evidence mostly small series; no proven outcome benefit.

Future

- Virtual and augmented reality models on the robotic console, point-of-care printing, drug-eluting printed stents.

Diagram to draw: CT → segmentation → STL → printer → colour kidney model with tumour, artery and calyces.

> **Key points to remember:**
>
> - Additive manufacturing from CT/MRI via segmentation and an STL file.
> - Main uses: partial nephrectomy and PCNL planning, consent, simulation.
> - Limits: cost, time, unrealistic tissue, weak outcome evidence.

Q3b. ECIRS (endoscopic combined intrarenal surgery)

Type: SAQ (10 marks)

Definition

- **ECIRS:** PCNL (percutaneous nephrolithotomy) and retrograde flexible ureteroscopy performed **at the same time, in one position**, by two surgeons.
- Term and technique popularised by **Scoffone (2008)** in the **GMSV (Galdakao-modified supine Valdivia) position**. Combined access dates back to the 1980s (Lawson's retrograde needle placement, 1983).

Rationale

- Each route covers the other's blind spots: the retrograde scope guides and watches the puncture, and reaches calyces the rigid nephroscope cannot.
- Result: one tract, better clearance.

Positioning

Position	Features
GMSV (preferred)	Supine; flank on a bolster at the table edge; ipsilateral leg extended, contralateral leg in lithotomy
Bart's modified supine	Variant of supine
Prone split-leg	Easier posterior calyces; needs a split-leg table

- **Supine advantages:** airway access, gravity drainage of fragments, no turning, simultaneous retrograde access.

Technique

1. Cystoscopy; UAS (ureteral access sheath) placed; flexible ureteroscope maps the stones and chooses the target calyx.
2. Ultrasound or fluoroscopic puncture **watched endoscopically** (double bull's-eye technique).
3. Wire grasped and brought out: through-and-through access.
4. Tract dilated under vision (balloon) and sheath placed.
5. Antegrade lithotripsy and suction; the retrograde scope **relocates stones** from inaccessible calyces into the nephroscope's view, or lasers them.
6. Final check with both scopes; stent or tubeless exit.

- Mini-ECIRS uses a mini-PCNL tract.

Indications

- Staghorn and complex multi-calyceal stones, combined renal and ureteric stones, anomalous kidneys, failed previous treatment, urinary diversion, obese or cardiopulmonary patients (supine).

Advantages (meta-analysis)

- Single position, **fewer tracts, less bleeding**, fewer ancillary procedures, **higher stone-free rate**; less fluoroscopy; the UAS keeps intrarenal pressure low.

Disadvantages

- Two surgeons and two sets of equipment; cost; longer set-up; ureteric injury from the sheath; less working space in supine.

Diagram to draw: GMSV position with nephroscope from the flank and flexible ureteroscope from below in the same calyx.

> **Key points to remember:**
>
> - Simultaneous PCNL plus flexible ureteroscopy in one position (Scoffone, GMSV).
> - Retrograde scope guides the puncture and relocates stones.
> - Fewer tracts, less bleeding, higher stone-free rate.

Q3c. Molecular targeted therapy in urological cancers

Type: SAQ (10 marks)

Concept

- **Targeted therapy** blocks a molecule the cancer depends on (receptor, kinase, transcription factor, DNA-repair enzyme, surface antigen), not all dividing cells.
- Classes: small-molecule TKIs (tyrosine kinase inhibitors, "-nib"), monoclonal antibodies ("-mab"), mTOR (mammalian target of rapamycin) inhibitors, HIF-2α (hypoxia-inducible factor-2α) inhibitor, PARP (poly-ADP-ribose polymerase) inhibitors, ADCs (antibody–drug conjugates), radioligands.

RCC (renal cell carcinoma)

- **Basis:** VHL (von Hippel–Lindau) loss → HIF accumulates even in normoxia → constant VEGF (vascular endothelial growth factor), PDGF (platelet-derived growth factor) and other targets.

Target	Drugs	Key evidence
VEGF receptors	Sunitinib, pazopanib, axitinib, cabozantinib (also MET, AXL), lenvatinib	COMPARZ: pazopanib noninferior to sunitinib; METEOR: cabozantinib beat everolimus
VEGF ligand	Bevacizumab	Modest PFS (progression-free survival) gain
mTOR	Temsirolimus (IV, poor risk), everolimus (oral)	Temsirolimus improved OS (overall survival) over interferon
HIF-2α	**Belzutifan**	VHL disease tumours; LITESPARK-005 beat everolimus after immunotherapy and TKI

- First line now: TKI **plus** checkpoint inhibitor. Everolimus also treats tuberous sclerosis angiomyolipoma.

Urothelial carcinoma

- **FGFR3 (fibroblast growth factor receptor 3)** alterations, typical of luminal tumours: **erdafitinib** (oral) after prior therapy.
- **ADCs: enfortumab vedotin** (Nectin-4) improved OS after platinum and immunotherapy; sacituzumab govitecan (Trop-2); HER2 (human epidermal growth factor receptor 2)-directed ADCs emerging.
- **EV-302 (2024):** enfortumab vedotin plus pembrolizumab, first line, roughly doubled OS versus platinum chemotherapy and is now standard (chapter notes predate it).

Prostate cancer (mCRPC, metastatic castration-resistant)

- **PARP inhibitors** exploit **synthetic lethality** in HRR (homologous recombination repair)-deficient tumours, especially BRCA2. **PROfound:** olaparib beat a second hormonal agent (imaging PFS 7.4 vs 3.6 months in BRCA/ATM).
- **PSMA (prostate-specific membrane antigen) radioligand: 177Lu-PSMA-617; VISION**: OS 15.3 vs 11.3 months.

Toxicity

- VEGF TKIs: hypertension, hand–foot syndrome, hypothyroidism. mTOR: mucositis, hyperglycaemia. Erdafitinib: hyperphosphataemia. Enfortumab: neuropathy, rash. PARP inhibitors: anaemia.

Limitations

- Resistance develops; cures rare; biomarker testing needed.

Diagram to draw: VHL–HIF–VEGF pathway with drug sites (TKI at receptor, bevacizumab at VEGF, mTOR and HIF-2α inhibitors).

> **Key points to remember:**
> - RCC: VEGF TKIs, mTOR inhibitors, belzutifan (HIF-2α).
> - Bladder: erdafitinib (FGFR3), enfortumab vedotin (Nectin-4).
> - Prostate: PARP inhibitors (BRCA2), 177Lu-PSMA (VISION).

Q3d. Artificial urinary bladder

Type: SAQ (10 marks)

Read as a man-made bladder substitute (prosthetic, engineered or transplanted). Bowel neobladder: see Practice Set 4 Paper 2, Q3e.

Need

- Bowel substitutes absorb urine and secrete mucus, causing acidosis, mucus, stones, infection, perforation and cancer. The aim is a substitute that **is bladder-like, not bowel**.

Requirements of an ideal substitute

- Low-pressure storage of adequate volume; continence.
- Complete, voluntary emptying; no reflux or upper-tract damage.
- Biocompatible: no encrustation, infection, rejection or cancer risk; durable.

Approaches

Approach	Principle	Status
Alloplastic (prosthetic) bladder	Synthetic reservoir (silicone, polymers) joined to ureters and urethra	Failed: encrustation, infection, leak at the tissue–prosthesis junction, migration, fibrosis
Tissue-engineered bladder	Autologous urothelial and muscle cells seeded on a biodegradable scaffold, omental wrap	Atala (2006): 7 patients improved; phase II trial showed no significant gain in capacity or compliance; experimental
Engineered conduit	Seeded scaffold as urinary conduit	Neo-Urinary Conduit trial: half developed stenosis or stricture; stiff, no blood supply
Bladder transplantation	Robotic retrieval and transplant of a vascularised composite bladder allograft	Feasible in models and brain-dead donors; first human case reported in 2025; needs immunosuppression

Why progress is slow

- **Vascularity:** thick constructs die centrally.
- **Innervation and contraction:** an engineered bladder stores urine but cannot yet empty voluntarily.
- **Urine–material interface:** foreign material encrusts and becomes infected.
- Fibrosis and contracture.

Alternatives in clinical use

- Augmentation (bowel, ureterocystoplasty, autoaugmentation); continent or orthotopic diversion; ileal conduit.
- **Also know:** neuroprosthetic emptying (sacral anterior root stimulator), artificial urinary sphincter, intraurethral valve-pump devices for underactive bladders.

Future

- Bioprinted vascularised constructs, stem cells (urine-derived, iPSCs, induced pluripotent stem cells), smart scaffolds.

Diagram to draw: biopsy → cell culture → seeding on bladder-shaped scaffold → omental wrap → anastomosis to bladder base.

Key points to remember:

- Prosthetic bladders failed (encrustation, infection, leak).
- Atala 2006 promising; later trial no gain; still experimental.
- Barriers: vascularity, innervation, urine–material interface.

Q3e. Artificial kidney

Type: SAQ (10 marks)

Read as devices that replace kidney function: the dialyser (classic "artificial kidney") and newer wearable, implantable and bioartificial kidneys.

Definition

- An **artificial kidney** removes solutes and water from blood across a semipermeable membrane. The haemodialysis dialyser is the standard example.
- Pioneered by **Kolff (1943)**; long-term use followed the Scribner shunt (1960) and Brescia–Cimino AV (arteriovenous) fistula (1966).

Principles

Process	Mechanism	Removes
Diffusion	Concentration gradient across membrane	Small solutes (urea, potassium)
Convection (haemofiltration)	Solute dragged with water	Middle molecules
Ultrafiltration	Hydrostatic pressure gradient	Water
Adsorption	Binding to membrane or sorbent	Some proteins, toxins

- **Hollow-fibre dialyser**; blood and dialysate flow **countercurrent** to keep the gradient.
- **Schedule:** blood flow 250–350 mL/min, 3–5 hours, 3–5 times a week; adequacy by Kt/V (urea clearance × time ÷ volume), at least 1.2 per session.
- **Peritoneal dialysis** uses the peritoneum as membrane (glucose or icodextrin draws water).

Limitations of current dialysis

- Intermittent: fluid and solute swings cause cramps.
- No endocrine (erythropoietin, vitamin D activation) or tubular metabolic function.
- Access thrombosis, infection, steal; cardiovascular disease, vascular calcification.
- **Transplantation remains the best renal replacement therapy.**

Newer artificial kidneys

Device	Principle	Status
WAK (wearable artificial kidney)	Belt-worn; sorbent regenerates a small volume of dialysate	Early human trials
Wearable or portable peritoneal dialysis	Continuous sorbent-regenerated peritoneal fluid	Development
Implantable bioartificial kidney	Silicon nanopore haemofilter plus bioreactor of renal tubule cells; driven by blood pressure, no pump	Preclinical
RAD (renal assist device)	Human proximal tubule cells in a hollow-fibre cartridge with continuous dialysis	Phase II in acute kidney injury; not adopted
Decellularised or organoid kidneys; pig kidney xenotransplant	Regenerative and xenograft routes	Experimental (see Practice Set 6 Paper 4, Q2)

Urologist's role

- AV fistula and graft work-up, laparoscopic PD (peritoneal dialysis) catheter placement with omental fixation, transplant.

Diagram to draw: hollow-fibre dialyser with countercurrent blood and dialysate flow.

> **Key points to remember:**
> - Diffusion, convection, ultrafiltration across a semipermeable membrane.
> - Dialysis lacks endocrine and tubular function; transplant is best.
> - Wearable, implantable bioartificial and RAD devices are investigational.

Q3f. Robotic kidney transplantation

Type: SAQ (10 marks)

Introduction

- **RAKT (robot-assisted kidney transplantation)**: the graft is implanted intracorporeally on the iliac vessels using the robot.
- Milestones: Hoznek (2002, robot-assisted anastomosis); Giulianotti (2010, obese recipient); **Menon and Ahlawat (Vattikuti–Medanta, 2014)** standardised the transperitoneal technique with **regional hypothermia**, assessed by the IDEAL (idea, development, exploration, assessment, long-term study) framework.

Indications and selection

- Living-donor grafts mainly; increasingly deceased donors.
- **Obese recipients** gain most, as open transplant carries more wound complications.
- **Contraindications (relative):** heavy iliac calcification on non-contrast CT (computed tomography), multiple graft vessels early in experience, previous transplant on that side, dense adhesions.

Technique (Vattikuti–Medanta)

1. Lithotomy, steep Trendelenburg; transperitoneal ports.
2. Periumbilical **GelPOINT** incision for graft and ice insertion.
3. Peritoneal flap; expose the **external iliac vessels**.
4. Graft wrapped in gauze with ice slush; **ice slush in the iliac fossa** for regional hypothermia.
5. **Vein first:** renal vein end-to-side to external iliac vein, to shorten arterial ischaemia.
6. Renal artery end-to-side to external iliac artery; unclamp and reperfuse.
7. Extravesical **Lich–Gregoir** ureteroneocystostomy over a double-J stent.
8. Graft retroperitonealised with the peritoneal flap to prevent torsion.

Advantages

- Small incision: **fewer wound infections, less lymphocele**, less pain, faster recovery, better cosmesis.
- Magnified 3D (three-dimensional) view and precise suturing in deep pelvis or obese patients.

Disadvantages

- **Longer rewarming (anastomosis) time**; cost; learning curve; pneumoperitoneum and Trendelenburg effects.
- No tactile feedback for calcified vessels.

Outcomes

- Multicentre and ERUS (European Association of Urology Robotic Urology Section) series: graft function and survival comparable with open transplant, with lower surgical-site complications.

Complications

- Graft thrombosis, haemorrhage, urine leak, ureteric stricture, lymphocele (less), port-site hernia; DGF (delayed graft function).

Diagram to draw: pelvis with graft on external iliac vessels and ureter to bladder dome; ice slush via GelPOINT.

Key points to remember:

- Vattikuti–Medanta technique with regional hypothermia (2014).
- Vein first, then artery, then Lich–Gregoir.
- Best for obese recipients; graft function equals open.

Rapid Revision Sheet – Practice Set 6 Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Present concept in management of hypospadias (LAQ)

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.

Q2. Discuss recent advances in tissue engineering with respect to urology (LAQ)

- Triad: cells, scaffold, signals; vascularity (100–200 µm diffusion) is the main barrier.
- Atala 2006 promising; phase II trial no gain; conduit trial half stenosed.
- Raya-Rivera 2011: seeded urethras in 5 boys.
- New: iPSCs, urine-derived cells, bioprinting, organoids.

Q3a. 3D printing in urology (SAQ)

- Additive manufacturing from CT/MRI via segmentation and an STL file.
- Main uses: partial nephrectomy and PCNL planning, consent, simulation.
- Limits: cost, time, unrealistic tissue, weak outcome evidence.

Q3b. ECIRS (endoscopic combined intrarenal surgery) (SAQ)

- Simultaneous PCNL plus flexible ureteroscopy in one position (Scoffone, GMSV).
- Retrograde scope guides the puncture and relocates stones.
- Fewer tracts, less bleeding, higher stone-free rate.

Q3c. Molecular targeted therapy in urological cancers (SAQ)

- RCC: VEGF TKIs, mTOR inhibitors, belzutifan (HIF-2α).
- Bladder: erdafitinib (FGFR3), enfortumab vedotin (Nectin-4).
- Prostate: PARP inhibitors (BRCA2), 177Lu-PSMA (VISION).

Q3d. Artificial urinary bladder (SAQ)

- Prosthetic bladders failed (encrustation, infection, leak).
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Q3e. Artificial kidney (SAQ)

- Diffusion, convection, ultrafiltration across a semipermeable membrane.
- Dialysis lacks endocrine and tubular function; transplant is best.
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Q3f. Robotic kidney transplantation (SAQ)

- Vattikuti–Medanta technique with regional hypothermia (2014).
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- Best for obese recipients; graft function equals open.